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Problem 3

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Classification Report:

	precision	recall	f1-score	support
Attack	1.00	1.00	1.00	11820
Normal	1.00	1.00	1.00	13375
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195

Confusion Matrix:

```
[[11809   11]
 [   33 13342]]
```

Fig. 1 Screenshot of Classification Report and Confusion Matrix

Problem 4

Classification Report for Dynamic Range Quantized Model:

	precision	recall	f1-score	support
Attack	1.00	1.00	1.00	11820
Normal	1.00	1.00	1.00	13375
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195

Confusion Matrix for Dynamic Range Quantized Model:

```
[[11809   11]
 [   33 13342]]
```

Fig. 2 Screenshot of Classification Report and Confusion Matrix for Dynamic Range Quantized Model

Problem 5

```
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Sample #1 | Predicted Class: 1 | Actual Class: 1 | Attack Score: 0.000000 | Normal Score: 0.996094
Sample #2 | Predicted Class: 1 | Actual Class: 1 | Attack Score: 0.000000 | Normal Score: 0.996094
Sample #3 | Predicted Class: 0 | Actual Class: 0 | Attack Score: 0.996094 | Normal Score: 0.000000
Sample #4 | Predicted Class: 0 | Actual Class: 0 | Attack Score: 0.996094 | Normal Score: 0.000000
Sample #5 | Predicted Class: 1 | Actual Class: 1 | Attack Score: 0.000000 | Normal Score: 0.996094
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```

Fig. 3 Screenshot of Serial Monitor showing output for first five test samples

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Sample #1 | Predicted Class: 1 | Actual Class: 1 | Attack Score: 0.000000 | Normal Score: 0.996094
Sample #2 | Predicted Class: 0 | Actual Class: 0 | Attack Score: 0.996094 | Normal Score: 0.000000
Sample #3 | Predicted Class: 1 | Actual Class: 1 | Attack Score: 0.000000 | Normal Score: 0.996094
Sample #4 | Predicted Class: 0 | Actual Class: 0 | Attack Score: 0.996094 | Normal Score: 0.000000
Sample #5 | Predicted Class: 1 | Actual Class: 1 | Attack Score: 0.000000 | Normal Score: 0.996094
Sample #6 | Predicted Class: 0 | Actual Class: 0 | Attack Score: 0.996094 | Normal Score: 0.000000
Sample #7 | Predicted Class: 0 | Actual Class: 0 | Attack Score: 0.996094 | Normal Score: 0.000000
Sample #8 | Predicted Class: 0 | Actual Class: 0 | Attack Score: 0.996094 | Normal Score: 0.000000
Sample #9 | Predicted Class: 0 | Actual Class: 0 | Attack Score: 0.996094 | Normal Score: 0.000000
Sample #10 | Predicted Class: 1 | Actual Class: 1 | Attack Score: 0.000000 | Normal Score: 0.996094
=====
```

Fig. 4 Screenshot of Serial Monitor showing output for ten new test samples